paper_id: PAPER_953
title: “Ramanujan-Accelerated S26 Convergence”
session: 214
date: 2026-04-12
author: “Daniel T. Murphy”
status: production
cvw: “v2.0.0”
tags: [buoyancy, 26D, UQFF]
sm_anchor: “CVW v2.0.0 — G6 SM Anchor Gate compliant”

PAPER_953: Ramanujan-Accelerated S26 Convergence

Author: Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 214 **Source:** spectral_{ladder_26state}.py (RamanujanAcceleration) **Calculator:** RamanujanAccelerationCalc (CP4 #537) **CVW:** v2.0.0 compliant

---

Abstract

We apply Ramanujan summation techniques to accelerate convergence of the 26-layer buoyancy sum $S_N = \sum_{k=1}^{N} \exp(-[\mathrm{SSq}] \cdot k/26)$. The accelerated form $S_N^{(R)} = S_N + \frac{1}{2}a_N + \sum_{p=1}^{P} \frac{B_{2p}}{(2p)!}\Delta^{2p-1}a_N$ uses Bernoulli numbers B_{2p} and forward differences to improve partial-sum accuracy at low N.

---

1. Ramanujan Correction

$$S_N^{(R)} = S_N + \frac{1}{2}a_N + \sum_{p=1}^{P} \frac{B_{2p}}{(2p)!}\Delta^{2p-1}a_N$$

where $a_k = \exp(-[\mathrm{SSq}] \cdot k/26)$ and $B_2 = 1/6$, $B_4 = -1/30$, $B_6 = 1/42$, $B_8 = -1/30$.

---

2. Convergence Comparison

N	S_N	$S_N^{(R)}$
5	partial	accelerated
10	partial	accelerated
26	exact S_{26}	$\approx S_{26}$

---

3. Source Data

- **File:** spectral_{ladder_26state}.py
- **Session:** 214
- **CP4 Class:** RamanujanAccelerationCalc (#537)

References

1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
2. Hardy, G.H. (1949) — Divergent Series (Oxford University Press)
3. Ramanujan, S. — Collected Papers (1927)
4. PAPER_952 — 26-State HRes Spectral Ladder
5. PAPER_959 — 26D Ramanujan Summation Engine
6. PAPER_960 — VDS Polylog26 Reference

Cross-Links

Related Paper	Relationship
PAPER_952	Spectral ladder that S_{26} accelerates
PAPER_959	Full 26D Ramanujan summation implementation
PAPER_960	Li_{26} cross-validation target
PAPER_949	BCS gap uses S_{26}

Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

> *Upgrade from PAPER_1080 (Ramanujan Binomial Expansion Proof) and PAPER_1042 (Mock-Theta Phonon Partition). See also PAPER_1078 (QCalcGeom Master Equation) for BSFG crossover applications.*

The third-order Ramanujan summation $S_{26}^{(3)}$, used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n\,(1/2)_n\,(3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + [\mathrm{SSq}] \cdot e^{-\kappa\, i\, n/26}\right]$$

where $(a)_n = a(a+1) \cdot s(a+n-1)$ is the Pochhammer symbol.

Binomial expansion (PAPER_1080): The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \qquad \text{with} \quad W_{26}(n) = \prod_{i=1}^{26} \left[1 + [\mathrm{SSq}] \cdot e^{-\kappa\, i\, n/26}\right]$$

This sum converges absolutely for $|[\mathrm{SSq}]| < 1$ (satisfied by $[\mathrm{SSq}] = 0.57$) and reduces to the classical Ramanujan $1/\pi$ series when $[\mathrm{SSq}] \to 0$.

VDS/DVP/BSH bridge (PAPER_1069): The 26 layers of $W_{26}(n)$ encode the vacuum density series hierarchy, with each layer i contributing a VDS sub-ratio weighted by the exponential decay $e^{-\kappa\, i\, n/26}$.

Mock-theta connection (PAPER_1042): The phonon partition function $Z_{\mathrm{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$ unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

Calibration Constants

Constant	Symbol	Value	Validation Domain
$[SSq]$	—	0.57	Polylog argument

Constant	Symbol	Value	Validation Domain
κ	—	$5.0 \times 10^{-4}\,\mathrm{day}^{-1}$	Damping rate
β_i	—	0.603	Buoyancy coupling
B_2, B_4, B_6, B_8	—	$1/6, -1/30, 1/42, -1/30$	Bernoulli coefficients

---

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
$S_{26}(0.57)$ convergence	≤ 50 terms to machine precision	Validated
Bernoulli coefficient identity	Standard number theory	Confirmed

---

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Mathematical Acceleration (Ramanujan Summation Methods)

§A.2 Generating Function

$$S_{26}^{(R)}(z) = \sum_{n=1}^{N} \frac{z^n}{n^{26}} \cdot R_n^{(26)} \xrightarrow{N\to\infty} \mathrm{Li}_{26}(z)$$

§A.3 Euler-Maclaurin Connection

$$\boxed{S_N^{(R)} = S_N + \sum_{k=1}^{p} \frac{B_{2k}}{(2k)!} f^{(2k-1)}(N)}$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 $\to$ VDS Li_{26} $\to$ Ramanujan $R_n^{(26)}$ acceleration $\to$ all phonon/jet/NS calculations

---

§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

S_{26} IS the VDS — it evaluates $\mathrm{Li}_{26}([SSq])$ directly.

§B.2 DVP

Prime factorization of convergence index maps to dipole vortex structure.

§B.3 BSH

Bernoulli coefficients provide the harmonic correction to buoyancy saturation.

§B.4 Production-Scale Consistency

Metric	Value	Status
Convergence terms	≤ 50	Confirmed
$[SSq]$	0.57	Confirmed

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1043	F_{U_Bi_i} Multi-System Buoyancy Curve Sweep
PAPER_1003	Spectral Ladder Merger 26-State Hierarchy
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation
PAPER_1080	Ramanujan Binomial Expansion Proof R_n^{(26,3)}

7 cross-reference(s) identified.